

Robotic Manipulator for Food Industry

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Abstract

Automation in the fruit-handling sector is steadily increasing due to various economic, operational, and labor-related factors. However, the development of robotic manipulators that can reliably handle delicate, irregularly shaped produce remains a complex challenge. To address this, a cost-effective robotic manipulator with a compliant soft gripper was developed using repurposed components from a 3D printer to reduce cost and complexity. A motion control glove, featuring various input sensors, was built to provide real-time control of the manipulator via wireless communication. The final design was capable of handling a range of fruits, such as apples and bananas, without causing damage, demonstrating the feasibility of low-cost robotic solutions for delicate object handling in agricultural automation.

Introduction

Automation has become a critical focus across agricultural and industrial sectors, driven by the need for greater productivity, quality control, and long-term sustainability. In the context of food production, particularly fruit handling, the pressure to improve efficiency and output has intensified due to global labor shortages and growing consumer demand for safe, high-quality produce. Automation offers a pathway to reduce dependence on manual labor while improving the consistency and safety of operations.

Despite advances in industrial robotics, manipulating soft, irregularly shaped objects like fruits remains a challenge. Fruits such as apples, bananas, and peaches vary in geometry and texture, making them prone to bruising. Traditional robotic systems often lack the adaptability and compliance needed for safe handling.

RAISE Engineering Consultants presents a low-cost robotic manipulator for agricultural automation. It is capable of handling delicate, irregularly shaped produce through real-time wireless operation using a sensor-integrated motion control glove.

Literature Review

A robotic manipulator is typically composed of two fundamental subsystems: the robot platform and the end-effector. The robot platform includes the structural framework, joint configuration, and actuators responsible for positioning and orienting the end-effector in space. The end-effector, meanwhile, is the functional component attached at the end of the manipulator, designed to interact with objects and perform task-specific operations. Careful consideration of both the robot platform configuration and end-effector design is essential to ensure effective operation.

Grasping in robotic systems typically follows three stages: approach, handle, and deliver [1]. The approach phase involves positioning the gripper near the object with the correct alignment to ensure a secure grasp. During the handling phase, the gripper is activated to lift and move the object as needed. Finally, the delivery phase focuses on accurately placing the object at the target location without causing damage or loss of control [1].

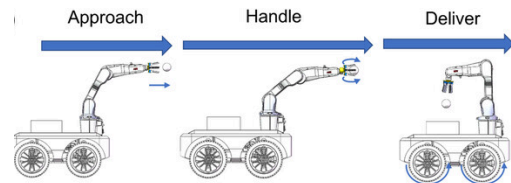


Figure 1 Gripper Grasping Stages [1]

Robotic grippers come in many forms and can be categorized in several ways, depending on various [2]. See [1, Fig. 2].

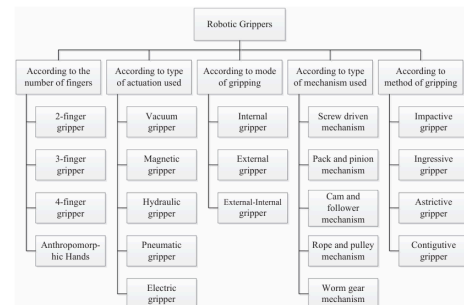


Figure 2 Gripper Classifications [2].

Mechanical grippers are the most common type, favored for their ability to handle objects of varying sizes and their compatibility with integrated force sensors [2]. Their adaptability has made them particularly popular in agricultural settings for handling delicate produce [2]. Conventional mechanical grippers are known for their simple and reliable operation, but their rigid structure and fixed degrees of freedom limited their ability to adapt to objects with variable shapes or softness [2].

Soft robotic grippers offer an effective solution for manipulating delicate agricultural and food products. They use highly deformable materials and operate through pneumatic or tendon-based actuation systems, which allow their structure to adapt to the shape of the object during grasping [1]. Moreover, their compliant nature lets them distribute forces evenly across the contact surface, which reduces the risk of bruising or mechanical damage [1]. They are also able to accommodate objects with varying shapes, sizes, and stiffness, which makes them ideal for agricultural environments [1], [2].

It is estimated that 60 to 70 percent of everyday objects can be effectively grasped with just two fingers [2]. While increasing the number of fingers can enhance performance and stability, it often requires more complex control systems [2]. For roughly spherical fruits, three-finger grippers are generally considered optimal [2].

Robot platforms, which define the structural and kinematic foundation of a manipulator, are typically classified into two main types: serial mechanisms and parallel mechanisms [3]. In serial mechanisms, the end-effector connects to the base through a series of links joined by single degree-of-freedom actuated joints, forming an open kinematic chain [3]. Parallel mechanisms, on the other hand, use multiple kinematic chains connected directly between the base and the end-effector, enabling coordinated movement through a closed-loop structure [3].

Cartesian robots, also known as gantry systems, operated using three prismatic joints that provide linear motion along the X, Y, and Z axes. They move in linear paths within a three-dimensional cubic work envelope, allowing for consistent accuracy and repeatable positioning [4]. These robots are ideal for tasks requiring straightforward linear motion, such as assembly, packaging, and pick-and-place operations [4]. However, their

fixed linear paths make them less flexible than articulated robots for navigating tight spaces or complex motions [5].

Articulated robots are composed of links connected by revolute joints [3], [5]. They offer a large and flexible work envelope, enabling complex motion paths and allowing them to reach into constrained or obstructed space[5]. These characteristics make them well suited for tasks such as assembly and pick-and-place operations [5]. However, their performance is limited by reduced stiffness and payload capacity, which affect their precision and suitability for heavy-duty applications [3], [5].

Selective Compliance Assembly Robot Arms (SCARA) consist of three revolute joints and one prismatic joint, allowing movement in a cylindrical work envelope that encircles their base [6]. SCARA robots are well-suited for high-speed, planar pick-and-place tasks due to their rigidity and horizontal flexibility [6]. However, their limited vertical mobility restricts their applicability in environments requiring significant height variation [6].

Delta robots are parallel manipulators composed of three identical arms connected to a fixed base and a moving end-effector [5], [7]. They are known for their high speed, stiffness, and positioning accuracy, making them ideal for lightweight pick-and-place tasks [5], [7]. Despite their advantages, Delta robots pose various challenges such as complex inverse kinematics and control algorithms, which require specialized expertise for proper programming and optimization [5], [7].

Cylindrical robots typically feature a revolute joint at the base for rotational movement, combined with prismatic joints that allow vertical and radial linear motion [5]. This configuration enables operation within a cylindrical work envelope, offering smooth and precise movement across circular areas and height levels [5]. The simplified joint arrangement reduces programming and control complexity, making these robots easier to design and build [5]. Cylindrical robots are well-suited for applications that require precise vertical and rotational movement, such as sorting, packaging, labeling, and inspection [5]. However, they present certain limitations, such as a restricted range of motion and relatively lower payload capacity compared to other industrial robot types [5].

Concept Design and Evaluation

The robot platform selected for this system was a cylindrical robot, chosen for its simplified joint configuration that enabled easier implementation by reducing both programming and control complexity. The mechanical design of the cylindrical robotic platform was divided into three main subassemblies: the base subassembly, the vertical motion subassembly, and the radial motion subassembly. These subassemblies were developed using a combination of custom CAD components and repurposed parts from a disassembled 3D printer, including stepper motors, rubber rollers, and T-slotted framing rails.

The base sub assembly facilitated the rotational movement of the entire structure. A stepper motor drove a 16-tooth motor gear, which meshed with a 100-tooth base gear attached to the central vertical column, achieving a 4:25 gear reduction for increased torque and angular precision. The rotation was supported by a component known as the Heart, which interfaced with six 45-degree tilted rubber rollers originally sourced from the 3D printer. These rollers acted as a thrust bearing to support the vertical load while ensuring smooth rotation. Additionally an assembly called the Shuriken, which housed four rubber rollers and acted as a radial bearing. These two components worked in tandem to secure the arm against any loads or moments generated by the assembly.

The vertical motion subassembly controlled the up-and-down movement of the radial axis. The structure was built using T-slotted framing rails from the repurposed 3D printer, providing rigidity and modularity. A stepper motor actuated a thread rod mechanism, also salvaged from the printer, which drove a 3D-printed gantry. This gantry assembly slid along the main column and carried the radial motion subassembly. Vertical motion was stabilized and guided using additional rubber rollers to reduce friction and maintain alignment.

The radial motion subassembly enabled horizontal extension and retraction of the end-effector. It featured a belt-driven carriage actuated by a stepper motor, mounted on another T-slotted framing rail. This carriage formed the mounting point for the gripper and allowed controlled radial displacement. The belt system offered a lightweight and effective solution for horizontal motion while maintaining speed and responsiveness.

The gripper used in this system was an off-the-shelf FESTO soft robotic design intended to gently conform to various object geometries [8]. Each of the three fingers is actuated independently by a servo motor mounted on a shared base, which is secured to the moving bracket of the radial axis. A four-bar linkage mechanism enables each finger to achieve approximately 40 degrees of motion.

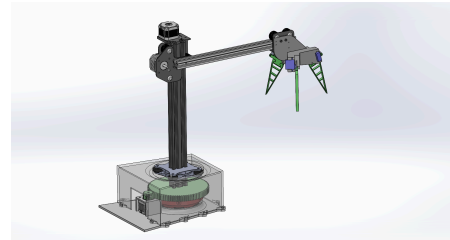


Figure 3 Designed Cylindrical Robotic Manipulator

The robotic manipulator is controlled by an Arduino Mega, which also interfaces with the motion control glove. Based on the incoming signals, the Arduino coordinates the movement of the joints using TMC2208 motor drivers. Creality 42-34 NEMA 17 stepper motors were repurposed from a Creality Ender 3 V2 3D printer. Additionally, the microcontroller controlled the gripper mechanism, which was actuated using SG90 servo motors. A PCA9685 driver was used to power the gripper mechanism.

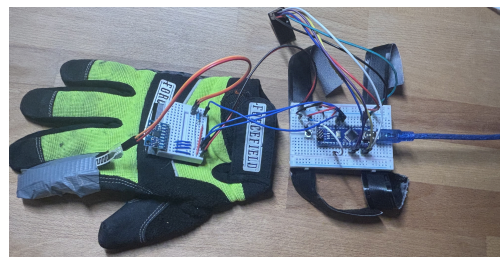


Figure 4 Motion Control Glove Prototype

The motion control glove was developed to provide wireless control of the robotic system. It featured a six-axis IMU to detect wrist orientation, a flex sensor to measure finger position, and a push button for discrete input. These sensors enabled control over base rotation, radial extension, vertical movement, and grip strength. Sensor data was processed by an onboard microcontroller and transmitted in real time to the robot using an NRF24 radio module, enabling hands-on operation of the cylindrical robot and supporting precision in fruit-handling tasks.

Results

The final robotic manipulator was successfully developed to operate in a cylindrical work envelope, offering controlled vertical, radial, and rotational movement with joint ranges of 24 cm, 26.5 cm, and 270 degrees, and corresponding precisions of 0.004 cm, 0.063 cm, and 0.288 degrees. The motion control glove transmitted user input wirelessly to the system at a frequency of 100 Hz, enabling responsive and continuous control during operation. The soft gripper, actuated by SG90 servo motors, was capable of exerting up to 0.821 kg of force per finger in approximately 0.204 seconds. It could accommodate spherical objects up to 93 mm in diameter and cylindrical objects up to 76 mm, allowing for secure grasping of various fruit types without causing mechanical damage.

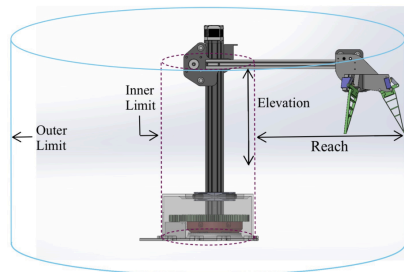


Figure 5 Manipulator Work Envelope

Conclusion and Recommendations

A cylindrical robotic manipulator was developed for handling delicate produce. The system combined repurposed and custom components to create an adaptable, low-cost solution capable of soft-object manipulation. The integration of a wireless control glove enabled real-time actuation and reliable handling across the manipulator's degrees of freedom. Future improvements may include implementing force sensors for autonomous grip adjustment, increasing joint speed, and incorporating trajectory planning and vision systems for fully automated control. The project demonstrated a functional and scalable approach to soft robotic manipulation in constrained, real-world environments.

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